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PROJECT NAME – SKILL-SCAN

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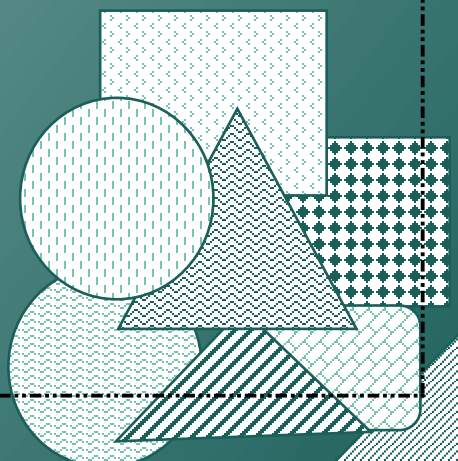
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Introduction

The recruitment process has become highly competitive due to the increasing number of job applicants and the growing demand for skilled professionals. Every organization receives hundreds or even thousands of resumes for a single job opening, making manual resume screening a time-consuming and challenging task. Recruiters often spend only a few seconds reviewing each resume before deciding whether it matches the job requirements. Therefore, candidates must prepare resumes that clearly highlight their technical skills, qualifications, and experience.

SkillScan is a web-based **Resume Analysis System** developed to simplify this process. The application allows users to upload their resumes in PDF format and automatically analyzes the content to identify technical skills mentioned in the document. Based on the extracted information, the system calculates an ATS (Applicant Tracking System) Score, predicts a suitable career role, identifies missing skills, and provides learning recommendations to help users improve their resumes.

The project is developed using the **MERN Stack (MongoDB, Express.js, React.js, and Node.js)**, which provides a fast, responsive, and scalable web application. MongoDB securely stores user information and analysis history, while React provides an interactive user interface. Express.js and Node.js manage the backend services, resume processing, and communication with the database.

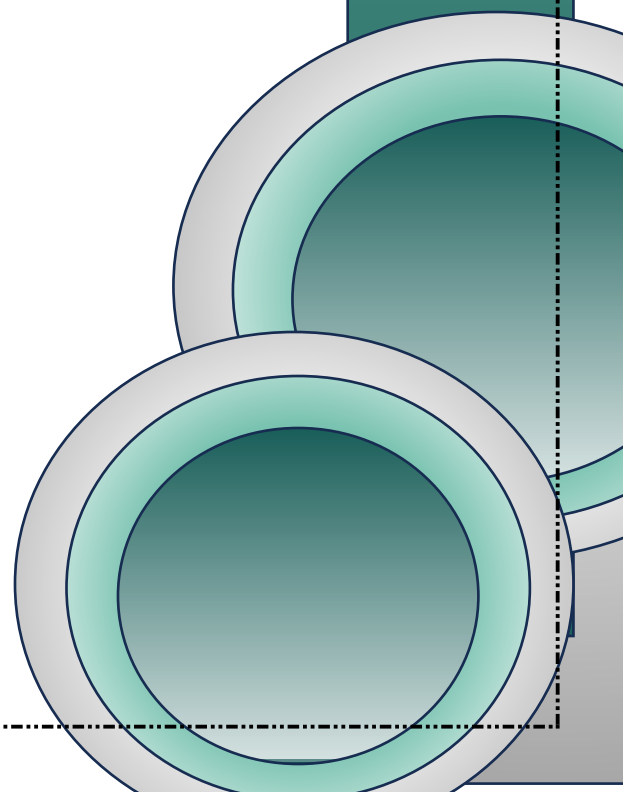
Unlike traditional resume builders, **SkillScan** focuses on resume evaluation rather than resume creation. It assists students, fresh graduates, and job seekers in understanding the strengths and weaknesses of their resumes before applying for jobs. Users can also view their previous resume analysis history, making it easier to track improvements over time.

The primary objective of SkillScan is to automate resume screening, provide meaningful insights, reduce manual effort, and help candidates prepare stronger resumes that increase their chances of getting shortlisted during recruitment.

OBJECTIVES OF THE PROJECT

The major objectives of the project are:

- To develop a web-based Resume Analysis System.
- To provide secure user registration and login functionality.
- To allow users to upload resumes in PDF format.
- To automatically extract skills from uploaded resumes.
- To calculate an ATS Score based on identified skills.
- To recommend suitable career roles according to the user's skills.
- To identify missing skills required for better job opportunities.
- To provide learning recommendations for improving technical knowledge.
- To maintain a history of previous resume analyses.
- To create a responsive and user-friendly interface using modern web technologies.



FEATURES OF SKILLSCAN

SkillScan provides various features to help users analyze and improve their resumes efficiently.

User Authentication

Provides secure user registration and login to access personal data.

Resume Upload

Allows users to upload resumes in PDF format for analysis.

Resume Analysis

Extracts technical skills and analyzes the uploaded resume.

ATS Score

Calculates an ATS score based on the detected skills in the resume.

Skill Detection

Displays all the technical skills identified from the resume.

Missing Skills

Highlights important skills that are not present in the resume.

Career Recommendation

Suggests a suitable career role based on the detected skills, such as:

- Frontend Developer
- MERN Stack Developer
- Java Developer
- Data Analyst
- Software Developer

Learning Recommendations

Provides suggestions to learn missing skills and improve employability.

Analysis History

Stores previous resume analyses for future reference.

Responsive Dashboard

Displays ATS score, skills, career recommendations, and analysis results through an interactive dashboard.

Development Tools

The following tools were used during the development of the project:

- Visual Studio Code – Source code editor
- Node.js – JavaScript runtime environment
- Express.js – Backend framework
- React.js – Frontend library
- MongoDB Atlas – Cloud database
- Postman – API testing
- Git & GitHub – Version control
- npm – Package manager
- Vite – React development environment
- Google Chrome- Browser Testing

FRONT-END TECHNOLOGY

The front-end of SkillScan is developed using modern web technologies to provide an attractive, responsive, and interactive user interface.

React.js

React.js is the primary JavaScript library used for developing the frontend of SkillScan. It allows the application to create reusable components, improving code maintainability and performance. React also provides faster rendering using the Virtual DOM, making the application highly responsive.

HTML5

HTML5 is used for designing the basic structure of web pages. It provides semantic elements that improve readability and accessibility.

CSS3

CSS3 is used for styling the entire application. Modern CSS features such as Flexbox, Grid Layout, Gradients, Glassmorphism, Hover Effects, Responsive Media Queries, and Animations are used to create an attractive user interface.

JavaScript

JavaScript is responsible for implementing client-side functionality such as form validation, event handling, API communication, and dynamic page updates.

React Router

React Router enables navigation between different pages like Dashboard, History, Career Insights, Login, Register, and Profile without reloading the webpage.

Axios

Axios is used to communicate with backend APIs and exchange data between the frontend and backend.

Chart.js

Chart.js is used for displaying graphical representations of resume analysis data such as ATS score and skill distribution.

React Icons

React Icons provides professional icons that improve the appearance and usability of the application.

BACK-END TECHNOLOGY

The backend of SkillScan is responsible for processing user requests, handling

Resume uploads, analyzing resume data, storing user information, and communicating

with the database.

Node.js

Node.js provides the runtime environment for executing JavaScript on the server side. It enables fast and efficient backend development.

Express.js

Express.js is used to build REST APIs that manage user authentication, resume uploads, resume analysis, and history management.

MongoDB

MongoDB is the NoSQL database used to store user accounts, resume analysis reports, ATS scores, career recommendations, and analysis history.

Mongoose

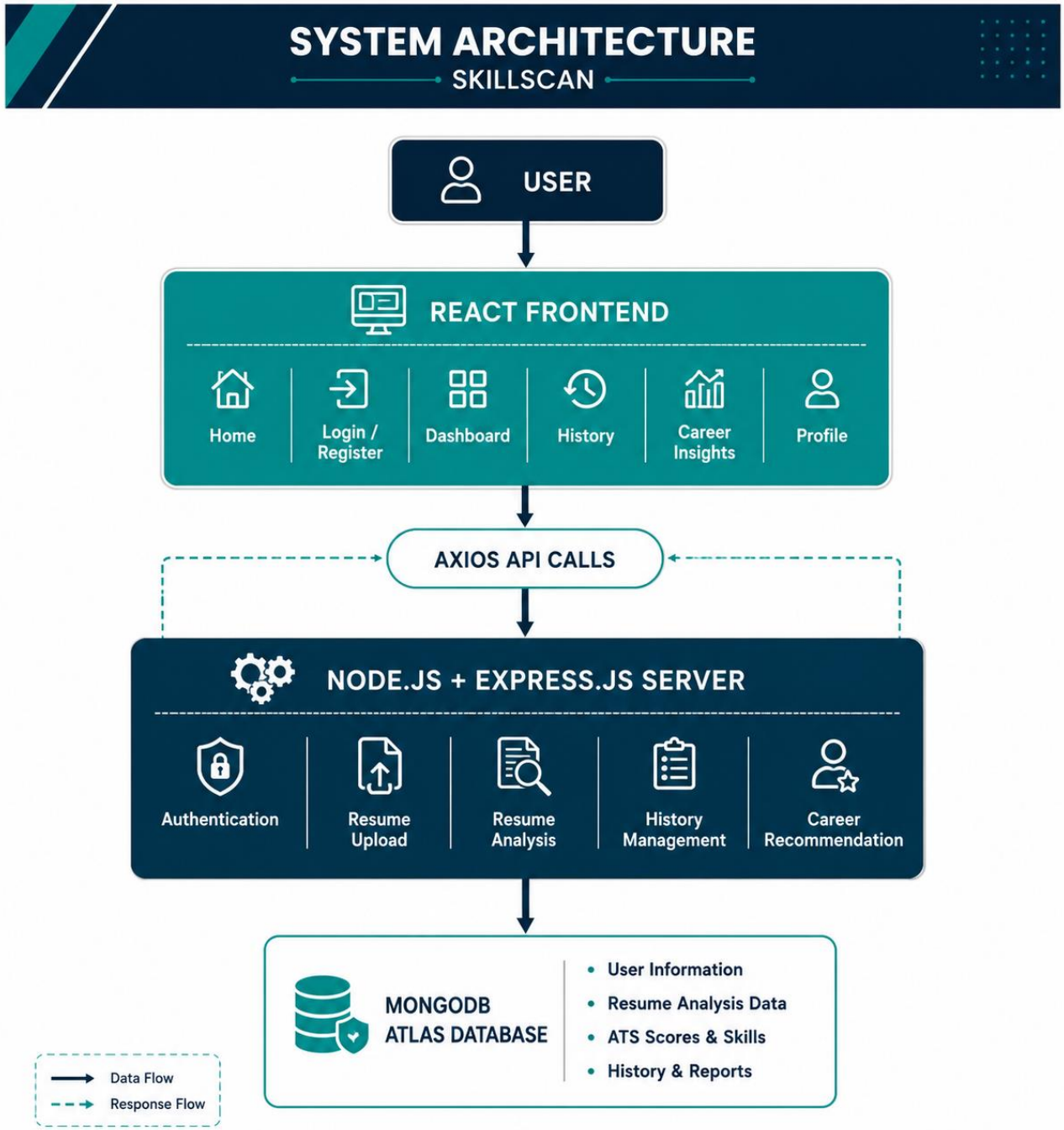
Mongoose is used for defining schemas, validating data, and interacting with MongoDB efficiently.

Multer

Multer is used for uploading PDF resumes from users to the server.

PDF Parser

The uploaded PDF resume is processed using a PDF parsing library that extracts text from the document. The extracted text is then analyzed to identify technical skills.


SYSTEM ARCHITECTURE
SKILLSCAN
USER
REACT FRONTEND
Home
Login / Register
Dashboard
History
Career Insights
Profile
AXIOS API CALLS
NODE.JS + EXPRESS.JS SERVER
Authentication
Resume Upload
Resume Analysis
History Management
Career Recommendation
MONGODB ATLAS DATABASE
User Information
Resume Analysis Data
ATS Scores & Skills
History & Reports
Data Flow
Response Flow

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